HAND & UPPER LIMB · NEET-SS / FET / INI-SS

Brachial plexus

1. Anatomy

- **Roots C5-T1** (variations: **prefixed C4-C8**, **postfixed C6-T2**). **Mnemonic: Roots, Trunks, Divisions, Cords, Branches** ("Randy Travis Drinks Cold Beer").
- **Roots** (between scalenus anterior and medius): branches **dorsal scapular nerve (C5: rhomboids, levator scapulae)**, **long thoracic nerve (C5-7: serratus anterior)**, contribution to the phrenic nerve (C5).
- **Trunks** (posterior triangle of the neck): **upper (C5-6)**, **middle (C7)**, **lower (C8-T1)**. Upper trunk branches: **suprascapular nerve (supraspinatus, infraspinatus)**, **nerve to subclavius**.
- **Divisions** (behind the clavicle): anterior and posterior; **no branches**.
- **Cords** (named by relation to the **axillary artery**, second part, deep to pectoralis minor):
  - **Lateral (C5-7): lateral pectoral nerve**, **musculocutaneous nerve**, lateral root of the median nerve.
  - **Medial (C8-T1): medial pectoral nerve**, **medial cutaneous nerves of arm and forearm**, **ulnar nerve**, medial root of the median nerve.
  - **Posterior (C5-T1): upper and lower subscapular nerves** (subscapularis, teres major), **thoracodorsal nerve (C6-8: latissimus dorsi)**, **axillary nerve (C5-6)**, **radial nerve (C5-T1)**.
- **Erb's point:** junction of C5 and C6 roots (upper trunk), **about 2-3 cm above the clavicle** behind the sternocleidomastoid.
- **Sympathetic chain: T1 contributes to the stellate ganglion: Horner syndrome (ptosis, miosis, anhidrosis, enophthalmos)** = **T1 preganglionic avulsion**.

2. Levels of injury: preganglionic vs postganglionic

Feature	Preganglionic (root avulsion; proximal to the dorsal root ganglion)	Postganglionic (distal to the DRG)
Recovery potential	**None (no spontaneous recovery, cannot be repaired directly)**	**Possible** (repair/graft, spontaneous)
Horner syndrome	**Present (T1)**	Absent
Proximal muscle weakness: **serratus anterior (long thoracic), rhomboids (dorsal scapular)**, **paraspinals**	**Paralysed** (winging)	**Spared**
Hemidiaphragm paralysis (phrenic, C5)	Possible	Absent
Tinel sign in the neck (supraclavicular)	**Absent**	**Present**
Severe neuropathic pain (burning, deafferentation)	**Common**	Less
SNAP (sensory nerve action potential)	**Preserved (despite anaesthetic skin)** (DRG cell bodies intact, peripheral axon intact)	**Absent**
Imaging (CT myelography/MRI)	**Pseudomeningocele**, absent rootlets	Normal roots
Histamine test (triple response)	Preserved	Absent
Fractures	Transverse process fractures of the cervical vertebrae, first rib fracture	Clavicle fracture

- **C5-C6 are most often postganglionic ruptures** (strong ligamentous attachment (transverse radicular ligaments) in the gutter of the transverse processes); **C8-T1 are most often avulsed (preganglionic)** (less protected).
- **Avulsion: central (intradural) avulsion** of rootlets; **peripheral avulsion** at the foramen.

3. Adult traumatic brachial plexus injury

- **Mechanism: high-energy traction**: **motorcycle accidents (young men)**: **head/neck forced away from a depressed shoulder** (upper plexus: C5-6) or **arm forced into abduction** (lower plexus: C8-T1).
- **Patterns:**
  - **Upper (C5-6) (Erb):** shoulder abduction/external rotation and elbow flexion lost; **"waiter's tip" posture**.
  - **Extended upper (C5-7):** + wrist and finger extension weakness.
  - **Lower (C8-T1) (Klumpke):** intrinsic hand paralysis, claw hand, ± **Horner** (rare in isolation in adults).
  - **Pan-plexus (C5-T1)**: commonest pattern in adults (**about 50-75%**): flail anaesthetic arm.
- **Associated injuries: subclavian/axillary artery injury**, **scapulothoracic dissociation** (**lateral displacement of the scapula on chest radiograph**, plexus avulsion + vascular injury: **poor prognosis**), **clavicle fracture**, **first rib fracture**, **head and cervical spine injuries**, **shoulder dislocation (infraclavicular plexus and axillary nerve)**.
- **Infraclavicular injuries** (cords/branches: shoulder dislocation, proximal humerus fractures) have **better prognosis** (postganglionic).
- **Investigations:**
  - **Chest radiograph** (hemidiaphragm, rib fractures), **cervical spine**.

- **MRI** or **CT myelography** (**pseudomeningoceles**: avulsion; CT myelography more sensitive for rootlet avulsion).
- **NCS/EMG at about 3-4 weeks** (SNAPs, paraspinal denervation: preganglionic).
- **Intraoperative** nerve stimulation, **somatosensory evoked potentials (SSEP)**, frozen section (ganglion cells: avulsion).

Management

- **Timing:**
  - **Open/sharp injuries: immediate exploration and repair**.
  - **Closed traction injuries with suspected avulsion (Horner sign, pseudomeningoceles): early exploration (within about 3 months)** for nerve transfers/grafts.
  - **Closed injuries with likely postganglionic lesions: observe 3 months**; explore if no clinical/EMG recovery (**ideally within 3-6 months**; **results deteriorate after about 6 months**; **motor end-plate degeneration by 12-18 months**).
  - **Vascular injury:** emergency repair (± plexus exploration).
- **Surgical options:**
  - **Neurolysis** (neuroma-in-continuity with conduction across it).
  - **Nerve grafting (sural nerve)** from viable roots (postganglionic).
  - **Nerve transfers (neurotisation):**
  - **Elbow flexion: Oberlin transfer** (**fascicle of the ulnar nerve to the biceps motor branch**), **double fascicular transfer** (ulnar fascicle to biceps + **median fascicle to brachialis**) (Mackinnon), **intercostal nerves to musculocutaneous nerve**, **spinal accessory to musculocutaneous**, **phrenic nerve**, **contralateral C7**.
  - **Shoulder: spinal accessory (distal) to suprascapular nerve** (via posterior approach preferred), **radial nerve (triceps long-head branch) to axillary nerve (Leechavengvongs/Somsak transfer)**.

- **Priority in pan-plexus injury: elbow flexion first**, then **shoulder stability/abduction**, then hand sensation, then wrist/finger function.
- **Free functioning muscle transfer (gracilis)** reinnervated by spinal accessory or intercostal nerves for late presentation or pan-plexus (elbow flexion and/or finger flexion: **Doi double free muscle**).
- **Tendon transfers** (incomplete injuries, late), **Steindler flexorplasty** (proximal advancement of the flexor-pronator origin for elbow flexion), **latissimus/pectoralis transfer for elbow flexion**, **trapezius transfer for shoulder abduction**.
- **Shoulder arthrodesis** (flail shoulder with **intact scapulothoracic muscles: trapezius, serratus anterior**): **about 20-30 degrees abduction, 20-30 degrees flexion, 40 degrees internal rotation** ("salute position"/hand-to-mouth).
- **Wrist arthrodesis**, **amputation** (above elbow with shoulder fusion: seldom improves function; **does not relieve pain**).

- **Neuropathic (deafferentation) pain:** pain medication (gabapentinoids), **dorsal root entry zone (DREZ) lesioning** for intractable pain in avulsions.

4. Obstetric (birth) brachial plexus palsy (BBPP/NBPP)

- **Incidence about 1-2 per 1,000 live births. Risk factors: shoulder dystocia (strongest)**, **macrosomia (large for gestational age, maternal diabetes)**, **breech delivery** (**bilateral, upper-root avulsions**), instrumental delivery, prolonged labour, previous affected child.
- **Narakas classification:**
  - **Group I: C5-6** (**Erb's palsy**; commonest; **best prognosis: about 90% spontaneous recovery**): **waiter's tip** (adducted, internally rotated shoulder, extended elbow, pronated forearm, flexed wrist).

  - **Group II: C5-7** (extended Erb): + wrist/finger extension loss (wrist drop).
  - **Group III: C5-T1 without Horner** (total palsy, flail arm).
  - **Group IV: C5-T1 with Horner syndrome** (**worst prognosis**).
  - (**Klumpke palsy, isolated C8-T1, is very rare.**)
- **Differential diagnosis (pseudoparalysis): clavicle fracture**, proximal humerus physeal fracture, **septic arthritis/osteomyelitis of the shoulder**, cerebral palsy.
- **Prognosis: recovery of biceps (elbow flexion) by 3 months** is the classic prognostic indicator (Gilbert and Tassin): **absence of biceps function at 3 months suggests poor spontaneous recovery: consider microsurgery (3-9 months)**. **Toronto Test Score** (elbow flexion, elbow extension, wrist, finger and thumb extension at 3 months; **score under 3.5 = surgery**), **cookie test at 9 months** (hand-to-mouth with the arm adducted). **Horner syndrome and total palsy: earlier surgery (about 3 months)**.
- **Management:**
  - **Physiotherapy (passive ROM: shoulder external rotation, abduction; prevent contracture)** from about 2-3 weeks.
  - **Primary microsurgical reconstruction** (neuroma excision + sural grafts, nerve transfers) at **3-9 months** if no biceps recovery.
  - **Assessment scales: Active Movement Scale (AMS)**, **Mallet classification** (shoulder function: abduction, external rotation, hand to neck, hand to back, hand to mouth; grades I-V).
- **Secondary deformities:**
  - **Internal rotation contracture of the shoulder** (**subscapularis/pectoralis major contracture, weak external rotators**): **glenohumeral dysplasia (posterior subluxation, glenoid retroversion, pseudoglenoid formation)**: **Waters classification (types I-VII)** on MRI/ultrasound. Monitor **passive**

external rotation (in adduction); **ultrasound screening** in infancy.

- **Treatment: botulinum toxin to subscapularis/pectoralis**, **open/arthroscopic release + reduction (subscapularis lengthening/release)**, **tendon transfer of latissimus dorsi and teres major to the rotator cuff (L'Episcopo/Hoffer/Sever)** for external rotation and abduction, **humeral external rotation (derotation) osteotomy** (late, with severe glenoid deformity).
- **Elbow flexion contracture**, **radial head dislocation (posterior)**, **forearm supination deformity** (biceps rerouting, forearm osteotomy), **"trumpet sign"** (shoulder abducts on attempted hand-to-mouth: weak external rotation).

5. Other brachial plexus conditions

- **Parsonage-Turner syndrome (neuralgic amyotrophy, acute brachial neuritis): sudden severe shoulder pain** (often after infection, vaccination, surgery) followed days later by **patchy weakness** (**long thoracic, suprascapular, AIN, axillary nerves** commonly); **pain resolves before weakness**; **hourglass constrictions** of nerve fascicles on high-resolution ultrasound/MRI. **Self-limiting** (recovery over months to years); steroids early; surgery (interfascicular neurolysis) for hourglass constrictions without recovery.
- **Radiation-induced plexopathy** (breast cancer; **myokymia** on EMG; usually **upper plexus**, painless) vs **tumour infiltration (Pancoast: lower plexus, painful, Horner)**.
- **Thoracic outlet syndrome** (see nerve compression notes).
- **Stingers/burners** (transient upper trunk traction or root compression in contact sport): unilateral burning dysaesthesia, transient weakness; return to play once symptoms resolve fully; recurrent: investigate (stenosis).

- **Brachial plexus tumours: schwannoma (commonest)**, **neurofibroma** (NF1, plexiform), **malignant peripheral nerve sheath tumour** (NF1; rapid growth, pain).

6. Examination checklist (rapid)

- **Proximal (preganglionic) clues: rhomboids** (dorsal scapular, C5), **serratus anterior** (long thoracic, C5-7: winging), **paraspinals** (posterior primary rami), **diaphragm** (phrenic), **Horner** (T1).
- **Trunk level: supraspinatus/infraspinatus** (suprascapular nerve: from the upper trunk).
- **Cord level: pectoralis major** (clavicular head: lateral cord (lateral pectoral); sternal head: medial cord (medial pectoral)), **latissimus dorsi** (thoracodorsal: posterior cord).
- **Terminal branches:** axillary (deltoid, regimental badge sensation), musculocutaneous (biceps, lateral forearm sensation), radial, median, ulnar.

> Exam pearl: **Preganglionic avulsion signs: Horner (T1), winging (long thoracic), rhomboid paralysis (dorsal scapular), paraspinal denervation, hemidiaphragm, absent Tinel, preserved SNAP with anaesthetic skin, pseudomeningocele. C5-6 rupture (postganglionic); C8-T1 avulse.**

> Exam pearl: **Adult closed plexus injury: explore by 3-6 months; motor end plates lost by 12-18 months. Elbow flexion priority: Oberlin (ulnar fascicle to biceps branch); shoulder: spinal accessory to suprascapular + triceps branch to axillary. Late: free functioning gracilis; flail shoulder: arthrodesis.**

Exam pearl: **Obstetric palsy: shoulder dystocia; Narakas I (C5-6) best, IV (with Horner) worst; no biceps recovery at 3 months (Gilbert) or Toronto score under 3.5 = microsurgery at 3-9 months. Secondary internal rotation contracture: glenoid dysplasia (Waters); latissimus/teres major transfer (L'Episcopo/Hoffer).**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.